

TrES-5b

R-1915

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190810 · created 2026-08-05T23:57:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458705.676 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.39 +/- 1.97 [%]
Orbital Inclination (inc)	82.52 +/- 2.89
Airmass coefficient 1 (a1)	0.8 +/- 0.47
Airmass coefficient 2 (a2)	0.15 +/- 1.24
Scatter in the residuals of the lightcurve fit is	57.39 %
Best Comparison Star	#3 - [244, 307]
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.052 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.054 vs 0.142 (deviation 0.46σ)
Duration fit vs published	0.052 d vs 0.0758 d (deviation 0.67σ)
Mid-time O–C	-2.9 min
Depth SNR	2.0
Residual scatter	57.39 %

Light curve

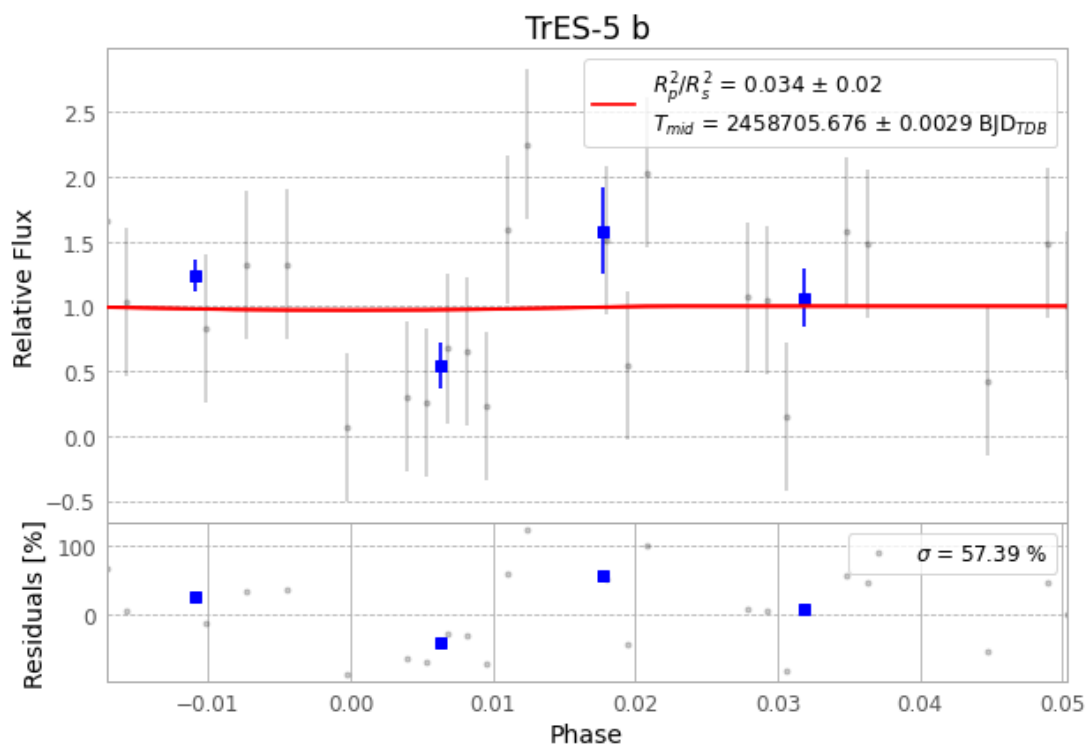


Figure 1 — FinalLightCurve_TrES-5 b_2019-08-09.png (SHA-256 6ea7c9e3ae19c9a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [471, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 955d75ab4bb9f7e8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

51 FITS frames (33 MB), TRES-5190810033315.FITS → TRES-5190810060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190810.log (f5a3d45929f9...), FinalLightCurve_TrES-5 b_2019-08-09.png (6ea7c9e3ae19...), FinalLightCurve_TrES-5 b_2019-08-09.csv (349f97d44cfb...)

Compute resources

Wall time 152.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:57Z.